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IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-SE-042

1. Topic (12 words max)

Machine Learning Based Smart Helmet for Personalized Construction Safety Monitoring

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	R26-SE-042
Year	2026

5. Brief description of the research problem including references (300 words max) – references not included in word count.

The construction industry plays a critical role in economic development; however, it remains one of the most hazardous industries globally, with high rates of accidents, injuries, and fatalities [1]. International Labour Statistics mentions that each year, around the world, around 60,000 fatal accidents occur on construction sites [2]. Traditional hazard identification approaches such as Job Safety Analysis (JSA), safety audits, and rule-based inspections are largely reactive, relying heavily on human judgment and predefined hazard lists [3]. These methods are time-consuming, fail to capture dynamic and unexpected risks, and often detect hazards only after incidents occur [4, 5, 6, 3]. This limitation has driven a shift toward proactive, data-driven safety approaches, including remote monitoring technologies.

Among remote monitoring technologies for hazard identification, Wearable Sensor Devices

(WSDs) have emerged as a promising solution due to their ability to provide real-time, individualized monitoring of environmental and physiological conditions [7, 8, 9]. Wearables are available in multiple forms such as helmets, wristbands, vests, and head-mounted devices, enabling continuous tracking of worker exposure to hazards. Despite their potential, these technologies are still in an early stage of adoption, particularly in real-world construction environments.

A key limitation identified in existing research is that most wearable systems operate using fixed threshold-based alert mechanisms. While thresholds provide a basic level of safety monitoring, they do not account for the complex and dynamic nature of construction environments or individual worker differences. This often results in false alarms, missed early warnings, and reduced reliability, limiting user trust and system effectiveness [7].

Furthermore, existing studies tend to analyze environmental hazards and worker physiological conditions independently, without considering their combined effect on safety [10]. This fragmented approach restricts the ability to capture real-world risk scenarios where multiple factors interact simultaneously.

Therefore, there is a need to investigate more advanced, integrated approaches that can analyze multi-dimensional safety data and support adaptive, context-aware decision-making for construction worker safety.

References

[1] P. Forsythe, "Proactive construction safety systems and the human factor," in *Institution of Civil Engineers*, 2014.

[2] International Labour Statistics, "Occupational Safety and Health Management in the Construction Sector," 2021. [Online]. Available: https://www.ilo.org/meetings-and-events/occupational-safety-and-health-management-construction-sector-3.

[3] H. Kim, H.-S. Lee, M. Park, B. Chung and S. Hwang, "Automated hazardous area identification using laborers' actual and optimal routes," in *Automation in Construction*, 2016.

[4] G. Carter and S. Smith, "Safety Hazard Identification on Construction Projects," in *Journal of Construction Engineering and Management*, 2006.

[5] Y. M. Goh and . D. K. H. Chua, "Case-Based Reasoning Approach to Construction Safety Hazard Identification: Adaptation and Utilization," in *Journal of Construction Engineering and Management*, 2009.

[6] H. S. Kim, H.-S. M. Lee, M. M. Park, B. Chung and S. S. Hwang, "Information Retrieval Framework for Hazard Identification in Construction," in *Journal of Computing in Civil Engineering*, 2013.

[7] I. Awolusi, E. Marks and M. Hallowell, "Wearable technology for personalized construction safety monitoring and trending: Review of applicable devices," in *Automation in Construction*, 2018.

[8] H. Chen, Y. Mao, Y. Xu and R. Wang, "The Impact of Wearable Devices on the Construction Safety of Building Workers: A Systematic Review," in *Sustainability*, 2023.

[9] H. Fugate and H. Alzraiee, "Quantitative analysis of construction labor acceptance of wearable sensing devices to enhance workers' safety," in *Results in Engineering*, 2023.

[10] I. Awolusi, C. Nnaji, E. Marks and M. Hallowell, "Enhancing Construction Safety Monitoring through the Application of Internet of Things and Wearable Sensing Devices: A Review," *ASCE International Conference on Computing in Civil Engineering 2019,* 2019.

[11] International Labour Statistics, 2023. [Online]. Available: https://www.ilo.org/resource/news/nearly-3-million-people-die-work-related-accidents-and-diseases.

6. Brief description of Existing Research and Systems (300 words max)

Existing Products and Systems

Recent advancements in construction safety have introduced a range of wearable sensor devices, including smart helmets, wristbands, smart vests, and body-mounted sensors. Smart helmet systems are among the most commonly implemented solutions, integrating sensors such as gas detectors, temperature sensors, and heart rate monitors to provide real-time safety monitoring and alert generation [1]. Similarly, wearable devices such as wristbands and smart garments have been used to monitor physiological parameters, worker activity, and fatigue levels, supporting continuous safety supervision on construction sites [2]. These systems typically transmit collected data to centralized platforms or dashboards for analysis and decision-making.

Existing Research

A substantial body of research has explored the application of wearable sensing technologies for improving construction safety. Studies highlight that wearable devices enable real-time monitoring of both environmental and physiological conditions, contributing to early hazard detection and improved situational awareness [3, 4]. Furthermore, recent research has begun incorporating artificial intelligence techniques to enhance safety monitoring, including fatigue detection, activity recognition, and predictive risk assessment [2]. These developments indicate a shift from traditional reactive safety approaches toward more proactive and data-driven systems.

Gaps and Limitations

Despite these advancements, several limitations persist in existing systems and research. First, many wearable safety solutions rely on fixed threshold-based alert mechanisms, which are insufficient for capturing the dynamic and complex nature of construction environments. Such approaches often lead to false alarms or missed critical conditions, reducing system reliability [1]. Second, existing studies frequently analyze environmental hazards and physiological conditions independently, without considering their combined influence on worker safety. This lack of integration limits the system's ability to represent

real-world risk scenarios where multiple factors interact simultaneously [5]. Third, there is limited implementation of personalized safety monitoring, as most systems apply generalized thresholds rather than adapting to individual worker baselines. These gaps highlight the need for integrated, intelligent, and adaptive safety monitoring systems in construction environments.

References

[1] I. Awolusi, E. Marks and M. Hallowell, "Wearable technology for personalized construction safety monitoring and trending: Review of applicable devices," in *Automation in Construction*, 2018.

[2] H. Chen, Y. Mao, Y. Xu and R. Wang, "The Impact of Wearable Devices on the Construction Safety of Building Workers: A Systematic Review," in *Sustainability*, 2023.

[3] P. S. Kumar, A. K. Thakur, K. Singh, S. Talari and S. Ali, "Smart Helmet - A Safety and Productivity Measures for Labourers in the Construction Sectors," *International Journal of Innovative Research of Science, Engineering and Technology,* vol. 14, no. 07, 2025.

[4] S. K. Shubhangi , A. A. Mayuri, G. S. Dattatray, J. S. Shivaji and S. V. Laxman, "IoT-Based Smart Helmet for Construction Worker Safety Using Raspberry Pi," *International Journal of Scientific Research & Engineering Trends,* vol. 11, no. 03, 2025.

[5] I. Awolusi, C. Nnaji, E. Marks and M. Hallowell, "Enhancing Construction Safety Monitoring through the Application of Internet of Things and Wearable Sensing Devices: A Review," *ASCE International Conference on Computing in Civil Engineering 2019,* 2019.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

Overview of the Proposed System

This research proposes an intelligent smart safety helmet system integrated with machine learning models and a web-based monitoring platform to improve construction worker safety. The system collects environmental and physiological sensor data from workers and analyzes it using machine learning to generate adaptive and personalized safety alerts.

Unlike existing systems that rely on simple threshold-based alerts, the proposed solution integrates two machine learning models that analyze environmental hazards and worker health conditions together. This allows the system to detect safety risks more accurately and provide early warnings to supervisors and workers.

Addressing the Identified Research Gaps

The proposed system addresses the identified limitations as follows:

1. Lack of intelligent analysis in existing systems: Many existing smart helmet systems rely on simple sensor monitoring and fixed safety thresholds. The proposed solution introduces machine learning models that analyze patterns in sensor data to predict potential safety risks.

2. Separate monitoring of environmental and physiological data: Current systems often analyze environmental hazards and worker health conditions independently. The proposed system integrates both types of data through a dual-model architecture, enabling more comprehensive safety assessment.

3. Lack of personalized safety monitoring: Existing solutions apply the same thresholds to all workers. The proposed system incorporates worker baseline data, allowing safety alerts to adapt to each worker’s physiological characteristics.

4. Limited intelligent decision systems: Instead of only transmitting sensor readings to dashboards, the proposed system generates automated risk assessments and alerts to assist site supervisors in making safety decisions.

System Perspective: Input – Process – Output

Inputs

The system collects the following data:

Environmental Inputs	Physiological Inputs	Worker Context Information
• Ambient temperature • Gas concentration levels • Noise level • UV radiation level	• Heart rate • Body temperature	• Worker ID • Historical baseline health data • Timestamp and working duration

Process

The system processes data through several stages.

Data Collection: Sensors embedded in the smart helmet collect environmental and physiological data in real time.
Data Transmission: The helmet sends sensor readings to the backend server via wireless communication.
Data Preprocessing: The backend system cleans and formats the sensor data.
Machine Learning Analysis: Two machine learning models perform safety analysis: • **Environmental Hazard Detection Model –** Analyzes environmental sensor data and identifies unsafe environmental conditions. • **Personalized Worker Safety Model -** Analyzes physiological sensor data and worker baselines and detects fatigue, abnormal health conditions, or stress.
Risk Fusion Layer: Combines outputs from both models to determine the final safety risk level.

Outputs

The system produces several outputs for construction safety monitoring.

Real-time Safety Risk Level	Automated Alerts	Safety Analytics
• Safe • Warning • Critical	Notifications sent to supervisors via the web dashboard	• Historical safety trends • Worker health and environmental monitoring reports.

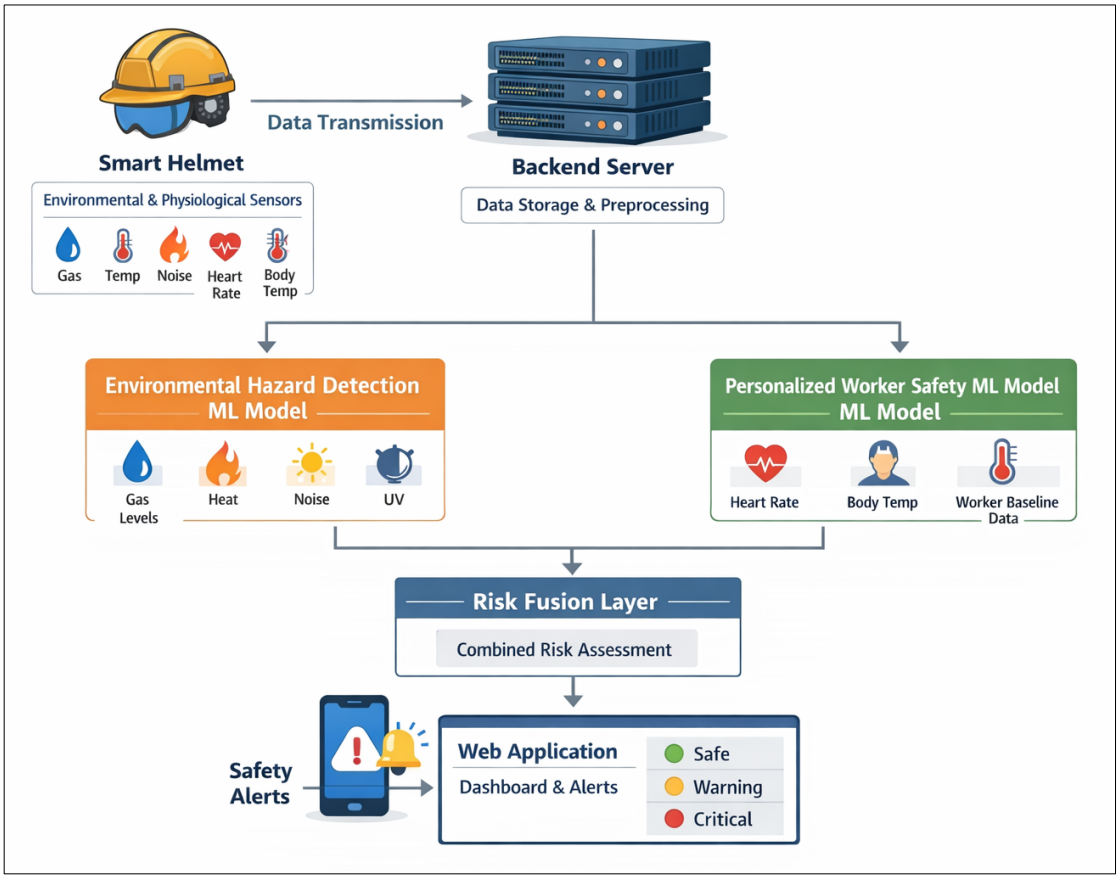


Figure 1: Conceptual Diagram

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

Access to Specialized Domain Expertise

The proposed research involves concepts from construction safety, wearable sensing technologies, and machine learning–based data analysis. Published research articles and conference papers from databases such as ScienceDirect, IEEE Xplore, ACM Digital Library, and MDPI will be used to understand existing wearable safety technologies and hazard detection systems in construction environments.

Industry knowledge can also be obtained through informal consultations with construction professionals, safety officers, or site supervisors who are familiar with occupational safety practices. Their insights can help to understand common environmental hazards, worker safety procedures, and practical deployment considerations for wearable safety devices.

Acquisition of Technical Knowledge and Skills

The technical knowledge required for this project includes embedded systems development, sensor integration, backend system development, web application development, and machine learning implementation. Can acquire these skills through academic coursework, online technical documentation, and open educational resources.

Sensor integration and hardware programming can be learned through development platforms such as Arduino or ESP32 microcontrollers, which provide extensive documentation and community resources. Machine learning techniques for safety prediction can be implemented using widely used frameworks such as Python-based libraries (e.g., Scikit-learn, TensorFlow, or PyTorch). Backend services and web dashboards can be developed using standard software engineering technologies such as Node.js, databases, and web frameworks. Tutorials, documentation, and open-source projects will support the learning process and practical implementation.

Data Collection for the Project

The data required for this research consists of environmental and physiological sensor readings. Environmental data include ambient temperature, gas concentration, noise level, and UV exposure, while physiological data include heart rate and body temperature.

Data will be collected through the prototype smart helmet during controlled experimental scenarios. Participants wearing the helmet will perform simulated construction-related tasks while sensor readings are recorded at regular intervals. The collected dataset will contain timestamped sensor measurements along with contextual information such as worker ID and task duration. This dataset will be used to train and evaluate the machine learning models responsible for hazard detection and personalized safety alerts.

Ethical Considerations

Since the project involves collecting physiological data from human participants, ethical considerations must be addressed. The study may require ethical clearance from the institutional research ethics committee before data collection begins. Participants will be informed about the purpose of the study and their consent will be obtained prior to data collection. Personal information will be anonymized, and collected data will be used only for research purposes. These measures ensure that the research complies with ethical standards related to human data collection and privacy protection.

9. Objectives and Novelty

Main Objective

To design and develop a machine learning–based smart helmet system that integrates environmental hazard detection and personalized worker safety monitoring to provide real-time, adaptive safety alerts for construction workers.

Member Name with Registration No	Sub Objective	Tasks	Novelty
IT21200892	Design and develop an intelligent smart helmet system integrating machine learning models for environmental hazard detection and personalized worker safety monitoring.	• Design and develop the smart helmet hardware with environmental and physiological sensors • Collect and preprocess sensor data for analysis • Develop two machine learning models for hazard detection and personalized safety monitoring • Implement backend system for data processing and model integration • Develop web application dashboard for real-time monitoring and alerts • Integrate all system components and perform testing and evaluation	Integration of dual machine learning models to combine environmental hazard detection and personalized worker safety monitoring within a single smart helmet system, providing adaptive and data-driven safety alerts instead of traditional threshold-based methods.

10. <u>This part is to be filled by the</u>

<u>Supervisor and the Co-supervisor of the Project.</u>

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes		No	

b) The proposed topic exhibits novelty.

Yes		No	

c) The student group can successfully execute the proposed project.

Yes		No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes		No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor	Dr.	Kalpani	Manuthunga	
Co-Supervisor	Ms.	Poojani	Gunathilake	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				